A Multimodal AI Agent Framework for Autism Behavioral Trait Detection in Children

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Abstract

This project presents a multimodal conversational AI agent for ASD behavioral trait detection in children through two complementary pathways: structured Q-CHAT-10 questionnaire responses classified by XGBoost, and caregiver descriptions classified by a fine-tuned BERT model with 14-category behavioural trait extraction. The system is orchestrated through a LangGraph multi-agent architecture: supervisor, questionnaire, text, story, schedule and guidance agents, where a large language model governs dialogue and routing exclusively, while all clinical predictions originate from trained ML models, eliminating hallucination by architectural design [2]. A persistent guidance agent provides personalised post-result conversation, explaining detected traits, concern levels, and recommended next steps. The XGBoost classifier achieves 98.6% accuracy on the Kaggle Toddler Autism Screening dataset. The BERT model achieves 88.0% accuracy on the Mendeley ASD Behavioural Text dataset. The system is deployed as a Streamlit web application, offering caregivers a parent-accessible first-level screening and guidance tool.

Introduction

ASD is a neurodevelopmental condition characterised by deficits in social communication, restricted interests, and repetitive behaviours with onset in early childhood [1]. The Q-CHAT-10 is a validated 10-item parent-report screening instrument for toddlers aged 12–36 months [3]. Despite growing interest in ML-based ASD detection, existing approaches share three critical gaps: most require structured clinical data unavailable to caregivers; LLM-based conversational tools risk generating fabricated clinical assessments [2]; and no prior work provides both screening and trait-level behavioural analysis within a single hallucination-controlled conversational agent. This work addresses all three gaps. The LLM is architecturally constrained to JSON routing output only and is blocked from generating any clinical result. All screening output, including 14-category trait detection, concern level scoring, and confidence-calibrated predictions, originates exclusively from trained ML models and a fixed clinical taxonomy.

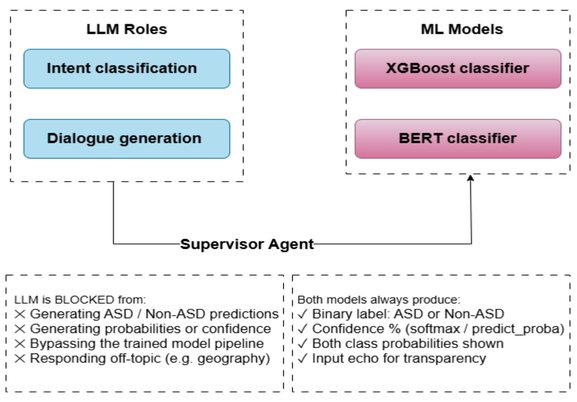


Figure 1 : What the LLM does vs what the ML models do

Proposed Methodology

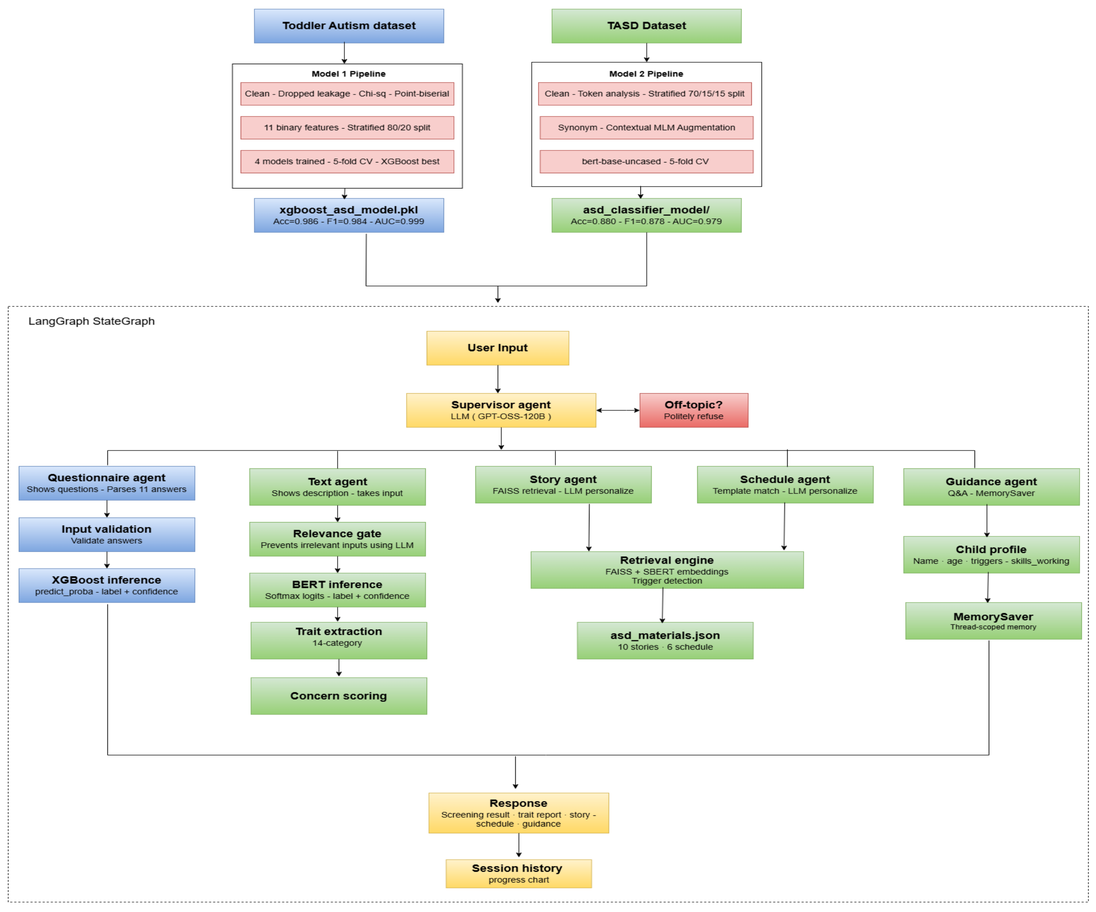


Figure 2 : Methodology

The agent is built as a LangGraph StateGraph with four nodes and a five-stage finite state machine. The supervisor agent reads the conversation stage and outputs constrained JSON routing decisions only, it never generates clinical content. The questionnaire agent presents the Q-CHAT-10 and routes answers to XGBoost inference. The text agent validates free-text input through a relevance gate, runs BERT inference, then calls the LLM-based trait extraction pipeline to identify which of the 14 behavioural traits are present and assigns a concern level (High / Moderate / Monitor) based on DSM-5 domain alignment. The guidance agent is a persistent conversational node that receives full assessment context — label, confidence, detected traits, and concern level — and answers caregiver follow-up questions across multiple turns using MemorySaver-backed persistent memory.

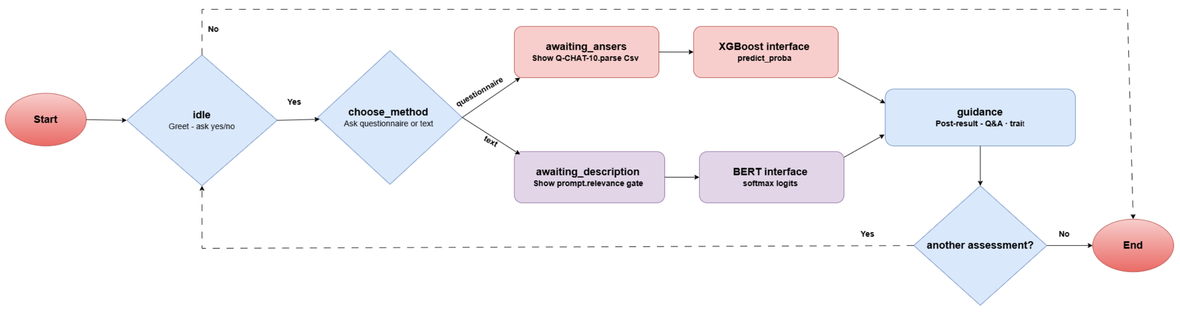


Figure 3 : ML Model prediction flow

Experiments & Result Analysis

Model 1 XGBoost: Four classifiers were trained on a stratified 80/20 split; XGBoost was selected via 5-fold cross-validation (CV=0.9763±0.014) for its highest accuracy and lowest variance.
Model 2 BERT: The Text dataset was split 70/15/15 before any analysis. Token length analysis on the training set justified MAX_LENGTH=64 with zero truncation. The training set was augmented using synonym replacement via WordNet with a 36-term protected clinical vocabulary, and contextual BERT MLM substitution, both applied to training data only.

Table 1: Model Comparison

Model	Accuracy	Precision	Recall	F1 (macro)	CV Accuracy
XGBoost	0.9858	0.9779	0.9897	0.9835	0.9763 ± 0.014
Random Forest	0.9621	0.9555	0.9555	0.9555	0.9597 ± 0.017
Logistic Reg.	0.9526	0.9413	0.9487	0.9449	0.9573 ± 0.026
KNN	0.9431	0.9259	0.9461	0.9349	0.9300 ± 0.030

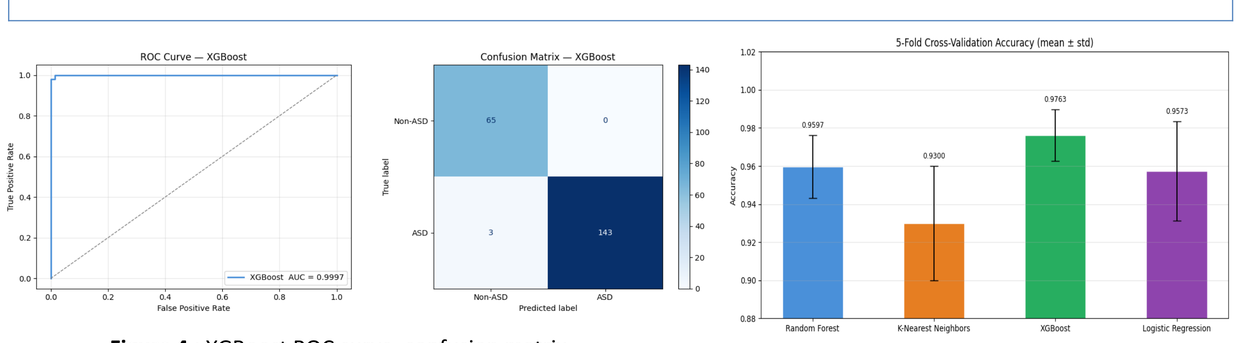


Figure 4 : XGBoost ROC curve, confusion matrix

Figure 5 : 5-fold CV results

Table 2: BERT Results

Metric	Score
Accuracy	0.8800 (95% CI: [0.762, 0.944])
Precision (macro)	0.8917
Recall (macro)	0.8766
F1 (macro)	0.8782
AUC-ROC	0.9792
5-Fold CV Accuracy	0.8825 ± 0.0423

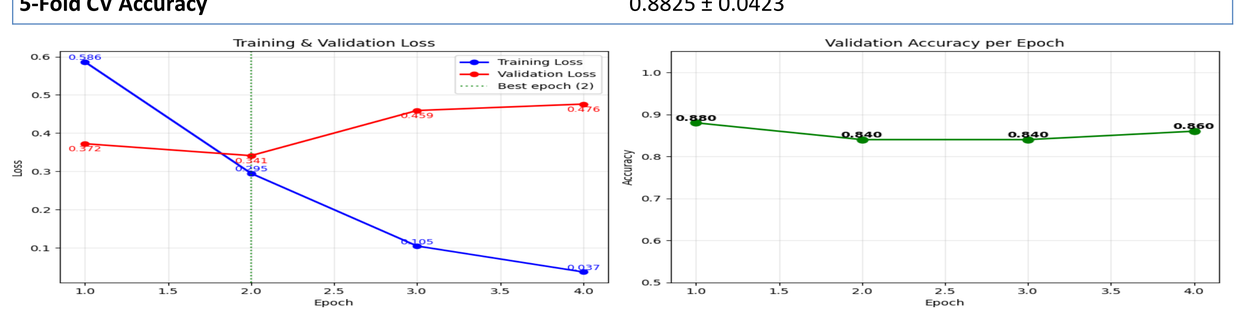


Figure 6 : BERT training curves

User Interface

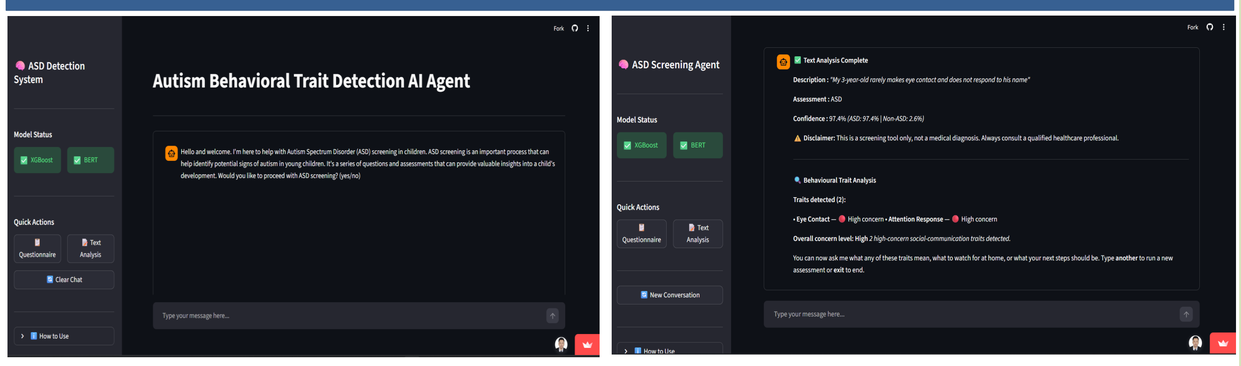


Figure 7 : Web Application

Discussion

The XGBoost classifier's near-perfect AUC-ROC (0.9997) and zero false positives confirm that the Q-CHAT-10 instrument provides strong discriminative signal when leakage variables are correctly excluded, consistent with prior ML screening studies [5]. The BERT model achieves an AUC-ROC of 0.9792 on behavioural descriptions, with CV accuracy (88.25%) closely matching held-out test accuracy (88.00%), indicating stable generalisation on a small dataset. The guidance agent extends the system beyond binary classification by providing trait-level behavioural analysis across 14 clinically grounded categories, mapping each to DSM-5 communication and restricted/repetitive behaviour domains. Persistent MemorySaver-backed memory enables coherent multi-turn caregiver conversations that maintain full assessment context across all turns. The strict LLM–ML architectural boundary prevents hallucination by design rather than prompt constraint, directly addressing the risk identified in LLM clinical applications [2]. Limitations include the absence of South Asian training data limiting regional generalisability, a small BERT test set (n=50, CI: [0.762, 0.944]), and the system's scope as a screening tool only, professional clinical assessment remains essential.

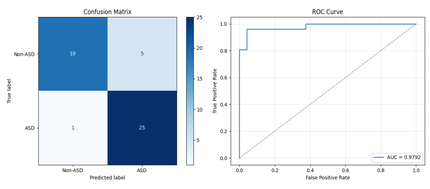


Figure 8 : BERT ROC curve, confusion matrix

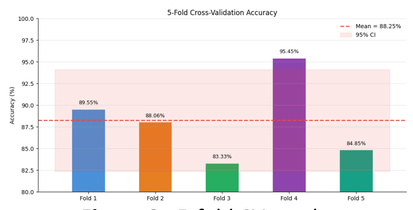


Figure 9 : 5-fold CV results

Conclusions

This work presents a parent-accessible, multimodal ASD behavioral trait screening and guidance agent combining structured Q-CHAT-10 questionnaire analysis (XGBoost: 98.6% accuracy, AUC=0.9997) with behavioural description analysis (BERT: 88.0% accuracy, AUC=0.9792, CV=88.25%±4.23%), 14-category trait extraction, concern level scoring, and persistent guided conversation within a hallucination-controlled LangGraph multi-agent architecture deployed as a Streamlit web application. The key contribution is the four-node agent design where the LLM manages dialogue while ML models and a fixed clinical taxonomy own all predictive and analytical output. Future directions include a larger region-specific behavioural text corpus, voice input for low-literacy caregivers, multilingual support in Bengali, and clinical usability evaluation with real caregivers.

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